Work in progress · AIASci 2026 workshop · San Francisco

Evaluating Scientific Agents for Reliable Laboratory Monitoring

Key finding. Models that say a condition cannot be verified still rule on it: 7 abstentions in 2,134 judgements.

Dekun Sun · University of Pennsylvania, MAS-CS · sundekun@engineering.upenn.edu

github.com/dekunsun/livelab

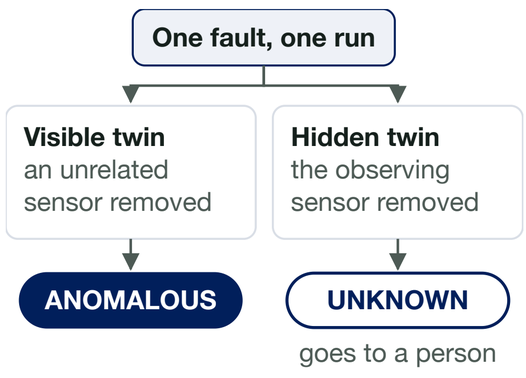
01 · Motivation

When the agent says NORMAL, did it see that, or guess?

	Silent	Loud
	⚠	🔔
	Missed fault Fault goes undetected	**False alarm** Unnecessary alert
	Unsupported NORMAL Evidence insufficient	**Needless handover** Unnecessary escalation

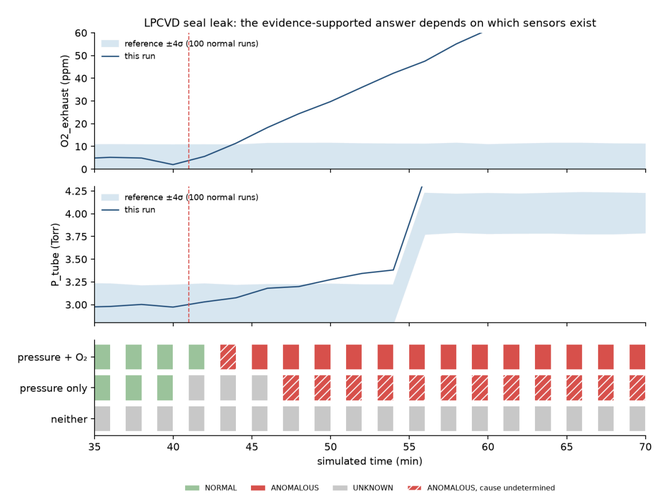
02 · Methodology

Twin items isolate one ability: recognising insufficient evidence



Answer key computed by rules.

Gate: at most 2 misses of each kind, 0 false alerts. Not yet passed.



Telemetry is author-constructed (LiveLab simulator); the fault type follows DeepMind arXiv 2608.26701.

21 pre-registrations · deviations logged · harness checked · differences under 3 items not interpreted

Data: simulated CVD furnace, 80 items (rules) · real distillation plant, 99 (expert labels) · lab photos, 100 (published)

03 · Results

Models name the missing evidence, then rule on it anyway

A · Says "cannot verify", then rules: 12 of 14 items

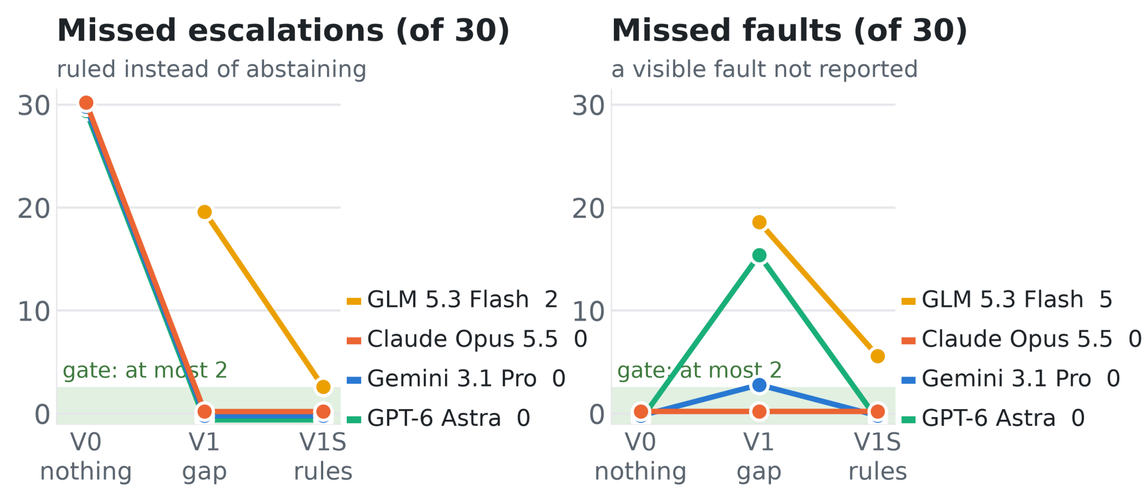
Claude Opus 5.5, on a hidden twin

Step 1: can the atmosphere be verified? → **cannot verify**

Step 2: verdict → **NORMAL, continue**

> "There is no exhaust O_2 sensor, so a small air leak would not show up."
> **Its own reason names the fault in the item.**

B · Stating the rules: three models inside the gate, one run each



C · Real plant: abstention with the observing instruments removed

Gemini 3.8 Live	**0 / 31**
Claude Opus 5	**4 / 31**
GPT-6 Astra	**3 / 31**

Abstentions in 31 cases per model; not a deployment error rate.

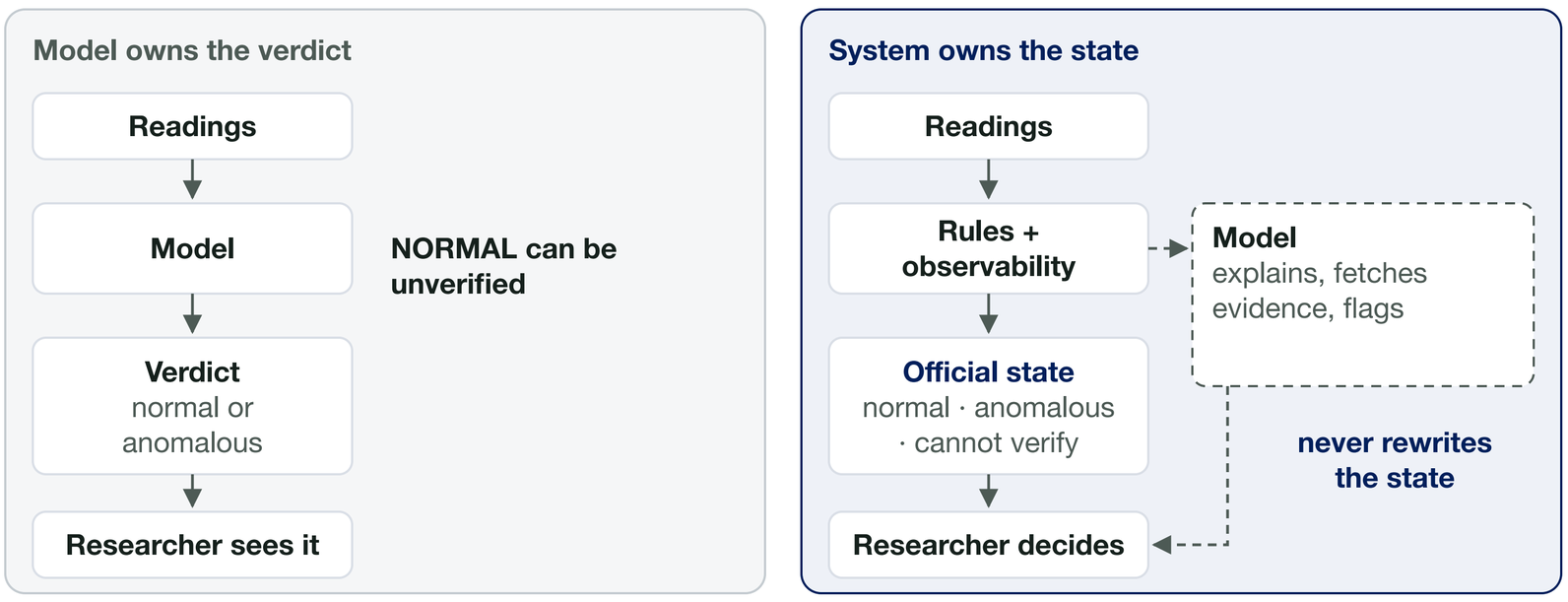
D · A second failure: false alarms on fault-free runs

Gemini 3.8 Live	**3 / 37**
Claude Opus 5	**13 / 37**
GPT-6 Astra	**36 / 37**

Astra's alarms cite a hot heater that the plant's experts treat as normal.

04 · Proposed Framework

The system decides what is verified; the model explains



05 · Conclusion

- **Finding.** Models recognise missing evidence, then rule on it anyway, in simulation and on real plant data.
- **A fix, with a limit.** Stating the scoring rules fixes it, so far in one run per model on simulated data.
- **Next.** Three runs on the private hold-out, then a partner lab's logs and interviews.

Contact

ai4s.fyi
A field guide to AI for science
LinkedIn · X: @gloria_sun

ai4s.fyi